School of Information and Physical Sciences

INFO6030 - Systems Analysis and Design

UON Fire Management System

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Lab Day: Wednesday

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Lab Time: 1:00pm- 3:00pm

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1. Introduction

1.1 Content

The University of Newcastle is in the process of undertaking an essential fire management system (FMS) upgrade across all the campus facilities. The FMS has the responsibility of assisting in the protection of the occupants and not only in the detection of a fire but its early suppression and compartmentation in addition to the process of evacuation in case of a fire outbreak. At the current setup, the FIP of each building is connected to a central unit also under the University's IFS through a computer network. This hub serves as a kind of a control center, to receive data from FIPs, send out orders, contain records of all the campus structures.

1.2 Objectives

The general aim of this project is to design an online management environment for IFS to regulate the FMS hub. It has to offer instant access to several important data to support the decisions made, have safe functional fire emergency response protocols, produce different sorts of reports, preserve security of records, and function continuously, day and night.

We aimed to:

1. Develop an effective interface that can be used on the World Wide Web to enable easy and effective operation of the University of Newcastle's FMS portal.
2. Ensure that all stakeholders receive immediate access to relevant information to respond effectively during fire emergencies.
3. Allow the system to produce assessment performance regularly, on the user's request, or at the end of an activity.
4. Make certain that the system meets the best practice guidelines and codes of practice in fire and emergency management.
5. Offer devices and features that enable the managers to make correct

decisions when there is a fire accident.

1.3 Requirements

In accordance with the guidance in 'Newcastle University - Infrastructure and Plant Services - Project Briefing Document - Fire Services Guidelines (October 2022, Version 1.3)', the design of the new system must meet the following key requirements:

- High reliability and safety: It is essential that the system is functioning continuously, without considerable downtime for maintenance, etc. to guarantee that the fire alarm and suppression system will operate as required under any circumstances.

- Real-time monitoring and data processing: The system should have the similar reception and data processing opportunities of real-time and should be capable of receiving/relying on information from different varieties of fire detection facilities and alarm systems (smoke detectors, temperature detectors etc.) in a particular time span.

- Diverse Reporting Functions: The system has to offer the availability of the following reports for managers which include the daily and the generated reports on need basis for efficient management and decision-making.

- Security and Rights Management: Another factor that must be implemented in the system is the need to protect data and records because it restricts access and only personnel with identifying clearance can access and work on it.

1.4 Major component

Fire management system is a huge system that integrates hardware and software. The system monitors fire conditions in the building through a central hub and multiple nodes connected to the central hub. Here are the main components of the system.

1. Central hub model:

A central management hub, operated by IFS, is connected to the FIP of each registered building. It can receive information and alerts from the FIP and issue remote commands. The central hub is the data center and control center of the entire system, which is responsible for monitoring the fire situation in the individual buildings, and at the presentation level, the central hub provides a real-time data display platform so that managers can observe the situation in the individual buildings.

2. Local fire indicator panel (FIP):

Each building has multiple FIP that take input data from various sensors and alarms within the building, as well as control alarm and fire suppression equipment. At the same time, FIP acts as the end node to upload data to the data center, which includes all operational records of the personnel using FIP and implementation

parameters of the sensors in the building.

3. Monitoring equipment:

Includes smoke detectors, temperature detectors, carbon monoxide detectors, manual pull alarms and manual break glass alarms. These devices are the end devices of the fire management system, and they pass the collected data to the FIP, which then passes the data to the data center.

4. Suppression system:

The suppression system is the first stage fire extinguishing system after the fire, which mainly includes the sprinkler on the ceiling of the building, carbon

dioxide generator and so on. The suppression system can control the fire's scale to a certain extent in its early stage. In the more complex fire environment, it is still necessary to rely on professional firefighters to extinguish the fire.

5. Evacuation system:

The function of the system is to control the equipment in the evacuation stage, mainly used to control the emergency lighting and control the power supply.

2. Business rules

2.1 Business rules of subsystem

Asset Management Subsystem:

1. Each new asset must include asset number, location, type, and status information.

2. Notification should be sent to the asset manager upon completion of asset registration.

3. All asset information updates must retain change history.

4. Asset information updates must be reviewed and approved by the asset manager.

5. The fire response team must have permission to view asset details.

6. Asset details must include maintenance history and current status.

7. Asset deletion must undergo dual approval from the asset manager and system administrator.

8. Deleted asset information must be retained for at least one year for audit purposes.

9. Asset status must be updated daily and displayed in real-time within the system.

10. Changes in asset status must automatically notify relevant personnel.

11. Asset maintenance tasks must be scheduled one week in advance and recorded on the system.

12. Each asset must be maintained at least once every six months.

13. Asset reports must include asset usage, status, and maintenance records.

14. Reports must be generated quarterly and sent to the asset manager and system administrator.

User Management Subsystem:

1. The publishing of new user records requires the intervention of the system administrator.

2. User account details must contain username, role, email address and telephone number.

3. Recursive changes must be capable of preserving the change history of the user information.

4. It was found that the updated information needed to be freely available in moments.

5. These accounts must be closed by the system administrator and the operations manager depending on the need.

6. PA data of deactivated accounts must be stored for a minimum of one year.

7. System administrators can view all user profiles of the members.

8. Profiles for users are maintained by the system, and users can view and edit their profiles.

9. It must be assigned by a system administrator because the role and

permission are predefined in the system.

10. It is also mandatory to have at least one role and permission for each user.

11. This means that basic user login would not be possible without the two-factor authentication.

12. A user must be locked out of the accounts after he enters three incorrect passwords, and this must be followed by an alert to the system administrator.

13. Frequent and important events should trigger messages to users who belong to the corresponding categories.

14. Notifications should consist of event details and suggestions for the action.

Event Management Subsystem:

1. Event records must include time, location, event description, and relevant personnel.

2. Event records must be completed within one hour of the event's occurrence.

3. Event updates must retain change history.

4. Event updates must be reviewed and approved by the system administrator.

5. System administrators and operations managers can view event logs.

6. Event logs must include detailed records of all relevant events.

7. Event assignment must be performed by the system administrator or operations manager.

8. Event priority must be determined based on urgency.

9. Relevant personnel must be immediately notified during an event.

10. Notifications must include event details and recommended actions.

11. Events must be prioritized based on urgency.

12. High-priority events must be handled immediately, with the results recorded.

Operation Management Subsystem:

1. Notice of maintenance must be made to anyone affected by it at least a week in advance.

2. It is necessary to ensure that maintenance tasks are registered into the system, and the relevant personnel is notified.

3. Any maintenance activity is to be captured on the system.

4. The movement of the maintenance schedule needs to be recorded and the concerned parties informed at the same time.

5. In the event of a fire, the fire response team must be called to respond to the incident on the spot.

6. It is also imperative to note that dispatch instructions should be documented in the system.

7. Responses to events must also provide information on all procedures followed whenever dealing with an event.

8. The response process must be logged into the system and those works must maintain history.

9. It is mandatory that the status of operations be updated and depicted in the system.

10. Another requirement of status updates is to inform relevant people.

11. Funding must be prioritized according to the concerns and exigency of the

event's context.

12. Recipients of the resources need to take and document the acknowledgement of the resources in the system.

13. It should be also noted that the team performance must be assessed and registered in the system continually.

14. Performance reviews should therefore be made more with emphasis on how effectively the event was handled and the time taken.

Reporting Subsystem:

1. It is necessary to submit the event reports containing descriptions of the event, the processes used to handle the event, and the results achieved.

2. Temporary reports must be submitted within one week of the event's resolution.

3. Specific aspects of maintenance must be documented in reports including tasks that have been done and the status and issues of the equipment.

4. There are required maintenance reports which must be prepared and presented monthly.

5. Status of assets must contain the current and historical data about all the assets at any specific time.

6. Such reports are required to be prepared and submitted monthly to the asset manager and system administrator besides being prepared and submitted quarterly.

2.2 Work Health and Safety rules

1. It is also required that all employees undergo section 508 workplace safety training to ensure compliance with accurate safety regulations and procedures.

2. It is mandatory for employees to put on personal protective equipment that include safety head gears including helmets, hand gears including gloves, protective glasses among others while engaging in activities that could be risky.

3. The workplace must have some form of emergency and evacuation procedure mapped out in detail and even have mock evacuations frequently to reduce the chances of employees being trapped in an emergency evacuation.

4. Every fire control item, including fire extinguishers alarm systems must be put under test and checked frequently to see that they are in good condition.

5. The workplace is required to undertake adequate hazard reconnaissance and employers risk assessment with proper precautionary measures put in place to prevent or otherwise minimize risks.

6. Any accident or 'close call' that occurs in the workplace must be immediately reported and may never go uninvestigated as this could lead to identification of the source of the potential problem that led to the occurrence of the accident, as well as steps taken to address the concern.

7. Any employee who might be subjected to hazardous materials or who is required to work under hazardous conditions shall be examined periodically for health risks.

He continues to work to the conditions put in place due to safety work requirements.

8. Employees and employers should also ensure that adequate safety measures are put in place in the workplace such as first aid kits and goggles.

other essential items required in emergencies such as other forms of medical equipment; guarantee that there are certified personnel to address any emergency that may occur around the camps.

9. Make sure that the employees spend adequate time off the workstation and have reasonable working hours to minimize on traumas incurred at the workplace besides any health complications.

10. Compliance with all specific WHS regulations within the context of the industry the undertaking is in, for example, WHS regulation on hazardous substances or the use of machinery (SafeWork NSW).

2.3 Ethical, Security, and Privacy rules

1. Data Confidentiality

- Do not disclose any personal information related to the building occupants. [Australian Privacy Principles]

2. Cybersecurity Measures

- Ensure the FMS is secure from hackers and other forms of cyber criminals.

- To provide a safe method for transmitting information between the hub and every FIP.

- Keep a log of all accesses, for any system or service, and all changes made to it. [ISO/IEC 27001]

3. Authorized Access Only

- Some suggestions which can work effectively are access should be limited to those who require it. [NIST Cybersecurity Framework]

4. Update Security Regularly

- The other fundamental aspect of leadership is to increase frequency of altering security measures.

- Education on security measures so that they will not violate these measures. [OWASP Top Ten]

5. Ethical Use of Tech

- Employ facial recognition and dual-gait facet ethically.

- It might be wise to take the ethical implications of the use of automated suppression systems into consideration. [AI Ethics Guidelines]

6. Data Transparency

- It is also important to inform occupants of the building where the testing is being carried out about their data being collected. [Australian Privacy Principles]

7. Follow Privacy Laws

- The data collection and management process shall obey to the set Australian Privacy Principles.

- Compliance with strict statutory requirements in data protection legislation. [Australian Privacy Principles]

8. System Audits

- Do this once per week or as often as is necessary to change them. [NIST Cybersecurity Framework]

9. Privacy in Reports

- Maintain personal identification of individuals in cases of emergencies where their information is reported. [Data Retention Guidelines]

10. Ethical Overrides

- Introduce procedures for constructive interaction regarding emergency

overrides.

- Emergency override identification needs to be supported by the appropriate documentation so that accountability for the action can be provided. [ISO/IEC 27001]

11. Command Protection

- The PLC and Fire Management circuit shall be programmed to prevent the sending of fire management commands from any unauthorized personnel. [OWASP Top Ten]

12. Data Retention Policies

- Ensure that it has clear policies as to how the data should be retained and when it must be erased. [Data Retention Guidelines]

13. Incident Response Plan

- Be prepared for such incidents or better still have a prepared way to handle security breaches.

Reference: The infrastructure will use the NIST Incident Response Guide to enhance the Incident Response Plan.

14. Regular Risk Assessments

- Risk adverse positions with respect to security and privacy should be constantly evaluated. [NIST Cybersecurity Framework]

2.4 Evidence of Research in Rules from Other Legislation and Relevant Standards

1. Adhering to the updated hazardous chemical labelling requirements as per the Globally Harmonized System (GHS) (SafeWork NSW).

2. Compliance with the licensing requirements for demolition work, including training and record-keeping (SafeWork NSW).

3. Implementing control measures for managing psychosocial risks in the workplace (SafeWork NSW).

4. Enhanced record-keeping and operator training requirements for amusement devices (SafeWork NSW).

5. Provision of digital induction training cards for general construction induction (SafeWork NSW).

6. Ensuring safe work environments as per the Work Health and Safety Regulation 2017 (SafeWork NSW).

7. Adherence to regular maintenance schedules for fire protection equipment (NSW Legislation).

8. Conducting regular fire drills as mandated by safety regulations (NSW· Legislation).

9. Proper disposal and management of hazardous waste materials in compliance with environmental regulations (NSW Legislation).

10. Provision and maintenance of adequate first aid facilities and trained personnel on-site (NSW Legislation).

11. NSW Work Health and Safety Regulation 2017 (NSW):

Part 10: A new regulation concerning fire prevention equipment and system – it ensures that all fire prevention equipment and system are checked, tested and maintained periodically in accordance with the requirement of relevant standard.

Section 74 Making sure that training and information on emergency procedures like fire safety and evacuation are provided to the personnel.

12. Environmental Planning and Assessment Regulation 2000 (NSW):

Clause 168: 168 provides that a fire safety schedule to determine the fire in the building is compulsory. This also includes the installation and maintenance of fire detectable and extinguishing apparatus.

Clause 170: Writing and submission of yearly fire safety statements are prescribed, mentioning that all fire safety measures are correct.

13. Building Code of Australia (BCA):

Section C outlines the fire resistance level with which building elements should be constructed to help curb fire spread.

Part 5 describes code requirements for fire detection, alarms, and extinguishers in different buildings, as explained in section E.

14.Australian Standard AS 1851:

Organizational procedures for frequent examination, evaluation, and maintenance of fire prevention systems are described to ensure the elements' preparedness to work at any given time.

Section 4 also presents frequency and methodologies for testing fire prevention apparatus including sprinklers, alarms, and extinguishers among others.

15. NSW Fire Brigades Regulation 2014:

Part 3: Regulates how building properties and tenants are supposed to undertake and perform their roles of ensuring that facilities have appropriate fire prevention measures and standards. This is combined with the first degree and contains such stipulations as ensuring the provision of access to the firefighting services and equipment.

Section 18: Explains the procedure to follow when communicating an upgrade of any fire protection systems or a change of activities that may affect fire safety in a building to the NSW Fire Brigades.

16. National Construction Code (NCC):

Volume One, Section C: Deals with fire protection and security, setting down the standards for fire rating for a building's elements to prevent or at least slow an escalation of fire.

Volume One, Section E: Outlines fire detection, alarm, and suppression systems in relation to different building classes and specifies the type of systems that are required under each class

17. Australian Standard AS 1670.1:

Fire Detection, Warning, Control and Intercom Systems - System Design, Installation, and Commissioning: Fire Alarm Systems – design, installation and commissioning of Fire Detection and Warning Systems, Fire Control Systems and Fire Intercom Systems.

Section 5: Defines the requirements of the indicated kind of detectors, their position in the building, and the interaction with other systems of building management.

18. NSW Environmental Planning and Assessment Act 1979:

Section 109R: Provisions requiring every fire protection measure should be kept in a way that corresponds to what the schedule on fire protection in the building expects.

Section 124: Assigns authority to the local councils to make orders for the remedial action and maintenance of facilities in terms of their fire safety matters if it is considered insufficient.

19.ISO 45001: Occupational Health and Safety Management Systems:

Clause 8.2: It means organizations need to develop Crisis Management Plans that cover all possible emergencies, such as fires and educate all individuals about what they should do in case of such an eventuality.

Clause 9.1 calls for the constant assessment of the emergency plans in that

process to confirm their effectiveness and then make necessary changes.

20. Privacy Act 1988 (Cth):

Part IIIA contains provisions on the control, storage and processing of personal information as far as occupants and movements of the buildings are concerned. It maintains the security and use of data solely for legal use and purposes.

Section 14 sets out the data security principles aimed at least personal information and providing that the enterprises ensure that the information is not misused, interfered with or lost.

21.Fire Safety Schedule (Environmental Planning and Assessment Act 1979):
Section 109R: Specifies that building owners need to have a fire safety schedule, with all fire safety files as well as their standard for performance on it and need to update this schedule frequently.

22. **Emergency Procedures (ISO 45001:2018 Occupational Health and Safety Management Systems)**:
Clause 8.2.3: They must adopt clear administrative measures for handling emergencies and test frequently, train the workforce and review the working of plans in realistic conditions.

3. Use Case Diagram

3.1 Use Case List

Asset Management Subsystem

Register New Asset: Record a new asset into the system with all the particulars must include location, type and status.

Update Asset Information: Update the specifics of an existing facility.

View Asset Details: Access an identified asset and obtain specific data regarding this asset.

Remove Asset: It also incurs the deletion of an asset from the system either totally or partially.

Track Asset Status: Supervise the conditions and statues of assets.

Schedule Asset Maintenance: Implement a schedule and plan and follow a proper time limit to perform an assessment on the assets.

Generate Asset Reports: Generate comprehensive documents of the current condition and previous use of assets along with the maintenance records.

User Management Subsystem

Create User Account: Create a new user under the new user section with the right roles and permission.

Update User Information: Update the user management and their roles and privileges.

Deactivate User Account: Suspend/terminate a user account either for a limited

time or for a long period.

View User Profile: Get profile information of a specific user.

Assign Roles and Permissions: Set clear user roles and access levels and responsibilities.

Authenticate User Login: Using password, ensure that users entering the sites authenticate themselves.

Notify Users: Notify or inform the users about something that may be of interest to them or some changes that they may need to make.

Event Management Subsystem

Log Event: It is a process of documenting a new event in the system.

Update Event Details: Update the elements of an event which is already scheduled.

View Event Log: Open and view a list of all the event recordings.

Assign Event to Team: Deploy a specific fire response team to a specific event to be ready in case a fire breakout happens.

Send Event Notifications: Notify associated staff about new activities and modifications.

Prioritize Events: Order events by importance; that is, split events into parts based on how severe they are.

Operation Management Subsystem

Schedule Maintenance: Holding the planned and scheduled maintenance of the assets.

Track Maintenance Activities: Track the current maintenance processes and their progress.

Dispatch Fire Response Team: Dispatch a team of firefighters to an emergency incidence location.

Manage Incident Response: Be responsible for an incident from investigation through to decision making and implementation.

Update Operation Status: Change the status of current operations and/or incidents.

Allocate Resources: Proper distribution of required resources to different operations and accidents.

Monitor Team Performance: Assess the activities of the response team during operations.

Reporting Subsystem

Generate Incident Report: It could be to write a report of a certain experience, a story, an event, or occurrence that took place on a university campus.

Generate Maintenance Report: Develop a document on maintenance activities.

Generate Asset Status Report: Make records containing the status of the assets.

Generate User Activity Report: Prepare the report of various activities carried out

by the user in the system.

Generate Compliance Report: Prepare a report which will enable you to meet safety and operational requirements.

Custom Report Generation: Users should be able to retrieve the information regarding the topic as per their requirements with details of the view.

Distribute Reports: Send generated reports to the party concerned based on the generated report and analysis.

Archive Reports: Store the reports for future use as the organization may need or wish to look at them sometimes.

Analyze Report Data: Conduct statistics on reports to comprehend peculiarities.

FIP (Fire Indicator Panel) Management Subsystem

Monitor FIP Status: The last precaution that must always be taken is to keep checking the status of the fire indicator panel.

Update FIP Configuration: Most of these configurations are typically found under the Administration tab as configuration settings of the FIP.

Log FIP Events: It can automatically log event summaries and alerts from the FIP in real time.

Generate FIP Report: They make reports out of the data and logs written by the FIP.

Perform FIP Diagnostics: Perform standard and non-standard tests on the FIP to check its activities.

Integrate FIP Data with Other Systems: Look at how FIP data can also be shared with other management and response systems.

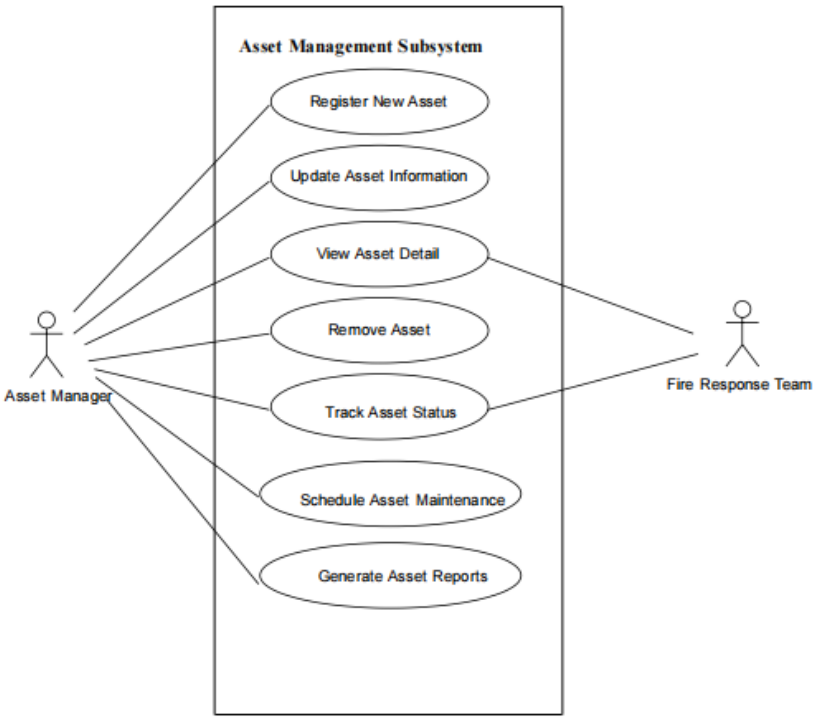
Schedule Regular FIP Maintenance: Development of a maintenance schedule of the FIP to avoid regular breakdowns.

3.2 Use Case Diagram

The use cases for all six subsystems have been diagrammatically represented below. The six subsystems in the use case diagrams interact with actors to perform various functions.

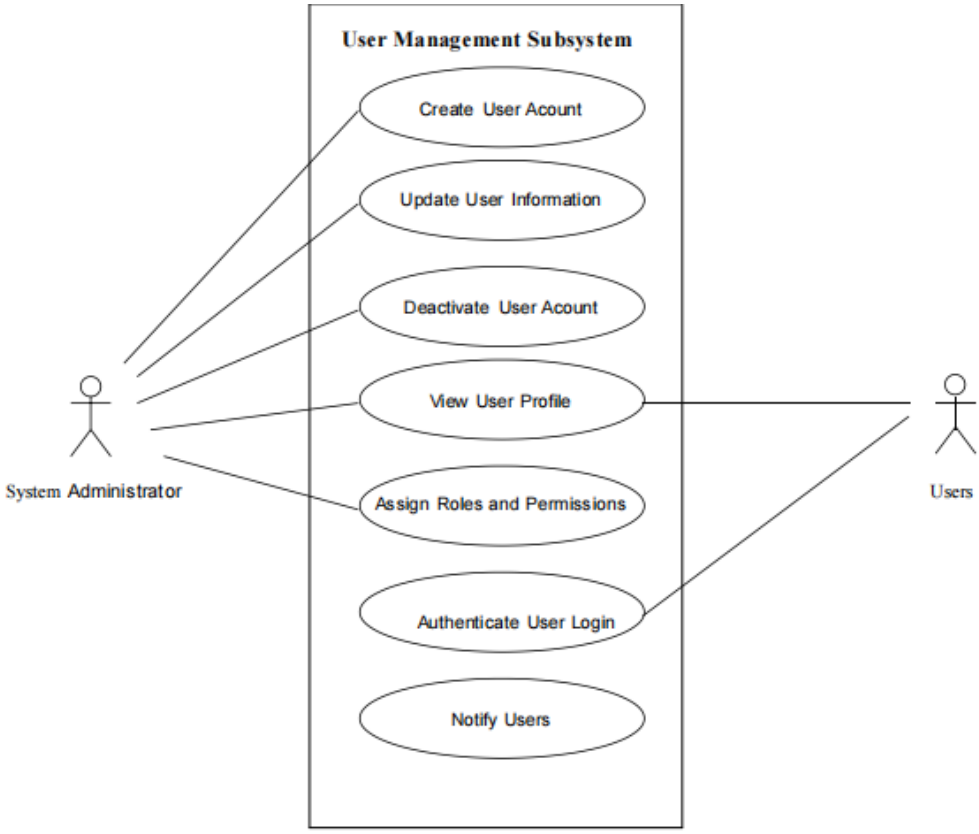
Asset Management Subsystem use case diagram

The Asset Management Subsystem allows actors to register new assets, update information, view details, track statuses, schedule maintenance, and generate reports, which this subsystem ensures that all assets can be maintained properly and to be ready for use during emergencies.



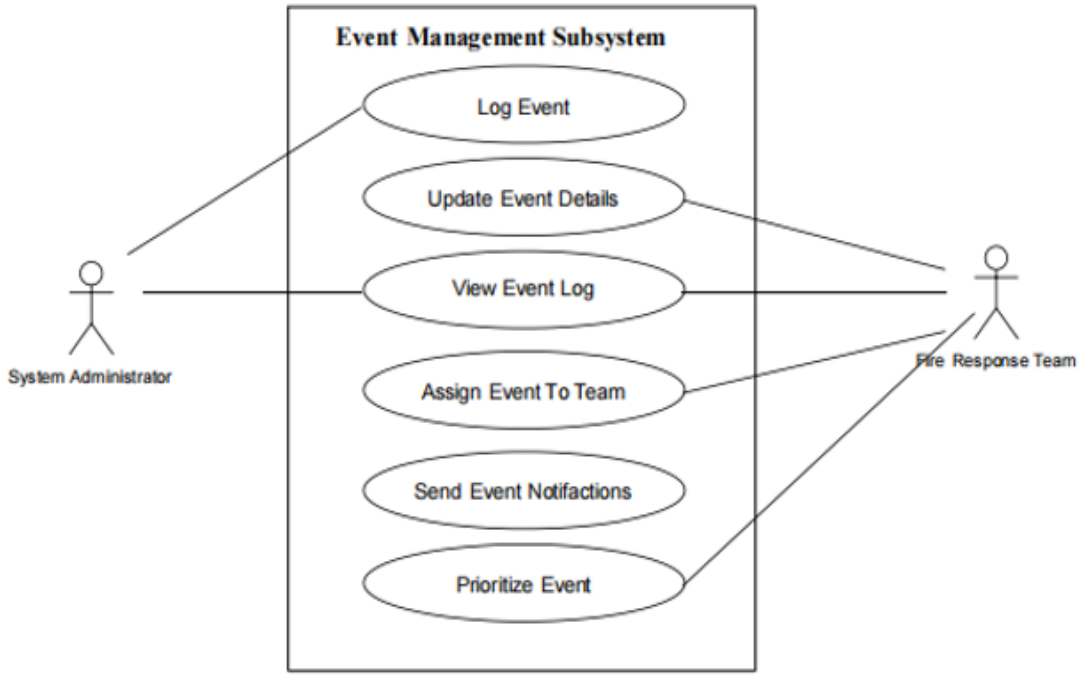
User Management Subsystem use case diagram

The User Management Subsystem allows actors to create and update user accounts, assign roles, authenticate logins, and notify users. This subsystem can ensure that only authorised person can access the system, and access to the private sensitive information and functions, so that the security of system can be maintained. Relative stakeholders can also view and update their information to ensure accurate data in database.



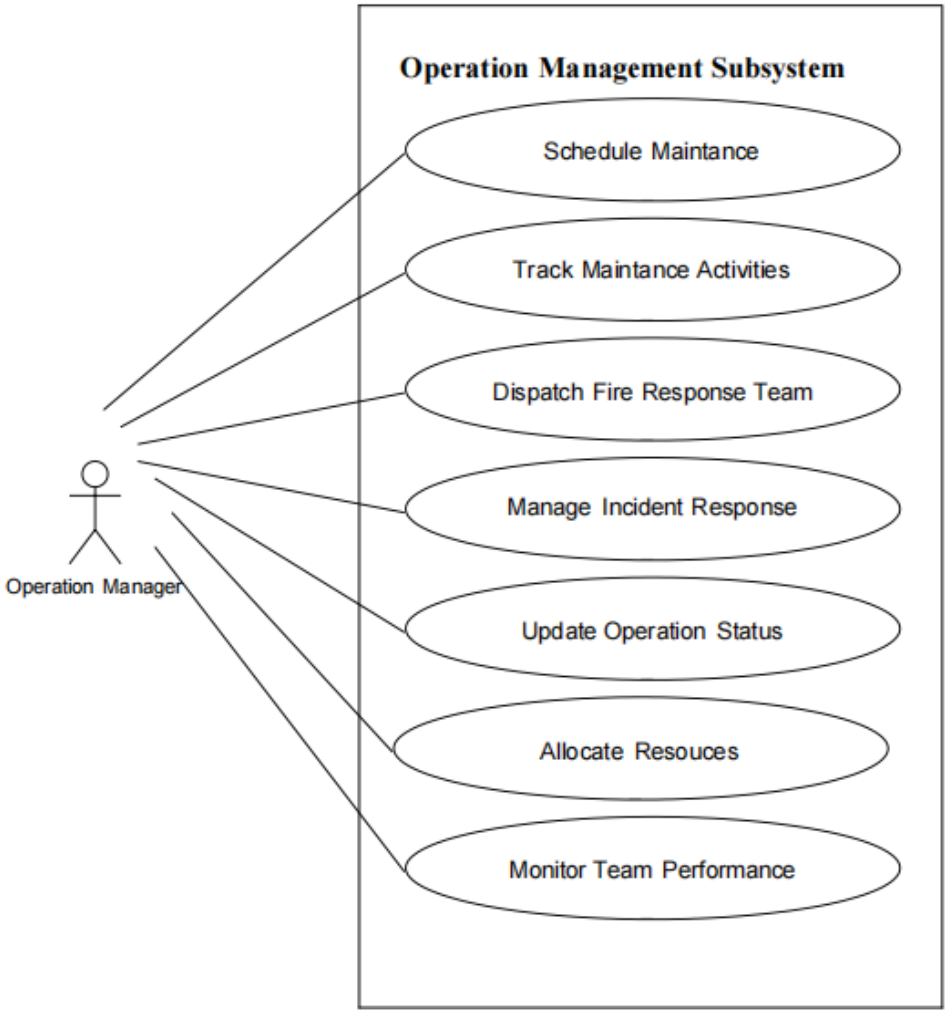
Event Management Subsystem use case diagram

The Event Management Subsystem enables actors to log events such as fire outbreaks, update details, view logs, assign responsible teams, send notifications, and prioritize events. This subsystem will allow relative stakeholders to react timely to emergencies that have been prioritised based on the severity of the incidents.



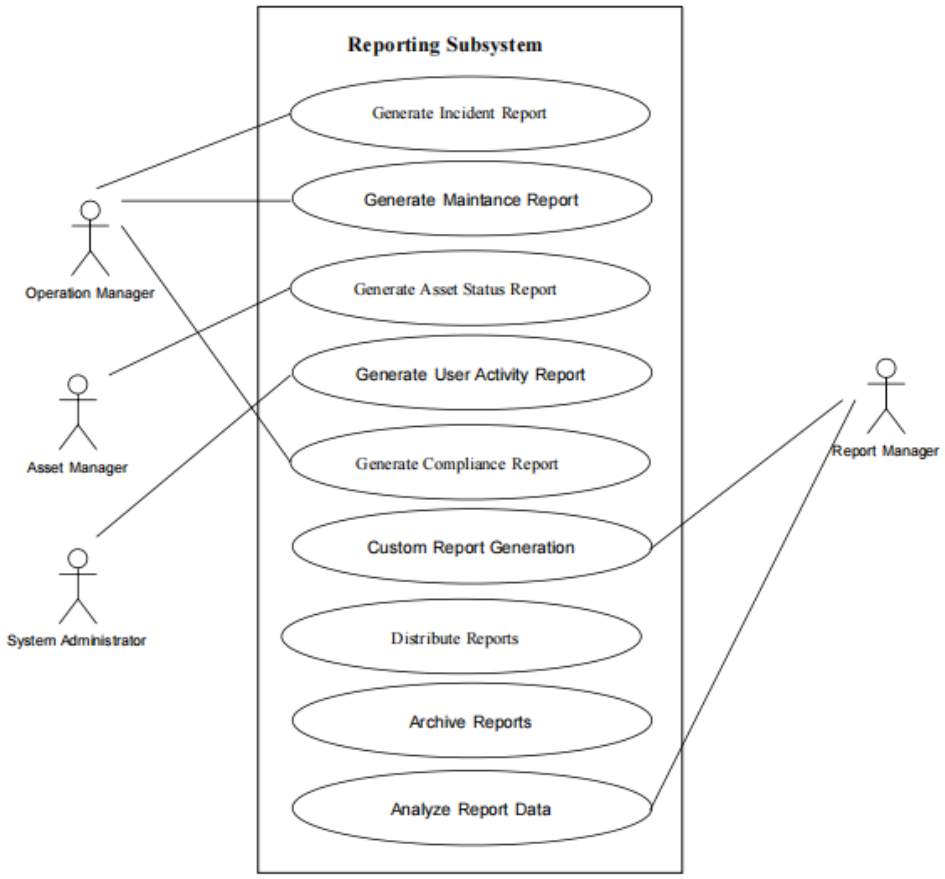
Operation Management Subsystem use case diagram

The Operation Management Subsystem allows operation managers to carry out scheduling and tracking of maintenance, dispatching fire response teams, managing incident responses, updating statuses, allocating resources, and monitoring team performance. Operation management subsystem ensures that all the operation activities are carried out smoothly and resources are allocated and utilised and efficiently and maintain that the incident management can be effective.


Operation Management Subsystem
Schedule Maintance
Track Maintance Activities
Dispatch Fire Response Team
Manage Incident Response
Update Operation Status
Allocate Resouces
Monitor Team Performance
Operation Manager

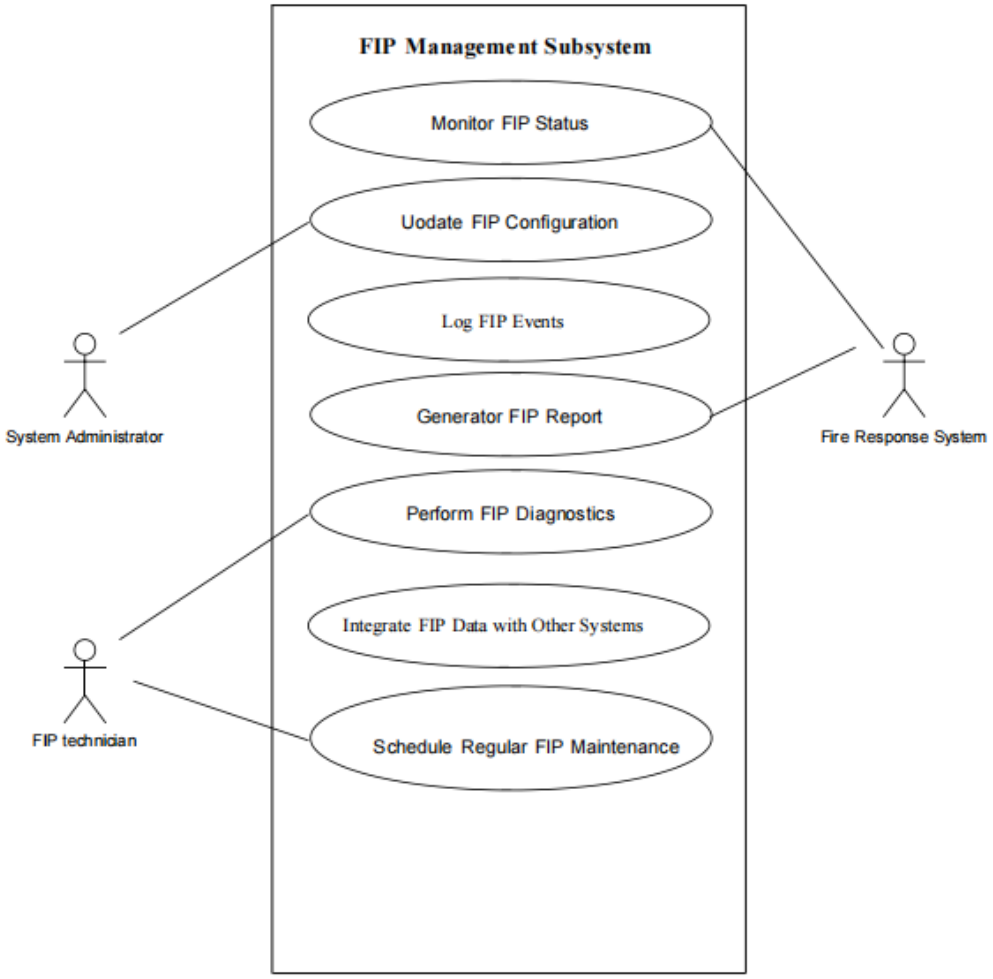
Reporting Subsystem use case diagram

The Reporting Subsystem enables actors to generate, distribute, and archive various reports, and analyse report data. The reports in the reporting subsystem will allow managers to evaluate the department's performance and ensure that all safety regulations and compliance have been followed. It also allows managers to make informed decisions and provide an all-rounded comprehensive overview of current operations, so that strategic planning can be performed accurately.



FIP Management subsystem use case diagram

The FIP Management Subsystem allows the technical staff to monitor and update the Fire Indicator Panel (FIP), log events, generate reports, perform diagnostics, integrate data, and schedule maintenance. This subsystem is very important as it can ensure the FIP can critically detect and alert fire incidents and ensure that all the precaution measures can function correctly while providing accurate data for managers.



3.3 Use Case Description

Table 1: Create User Account description (Kun Zhang)

Use Case Name:	Create user account	
Scenario:	Create online user account in the FMS	
Triggering Event:	Administrator wants to set up account for a new user	
Brief Description:	The administrator creates accounts for members who need to join the fire management system. These members can use the accounts to log in to the system and perform operations on the system within their respective permissions.	
Actors:	System Administrator	
Related use cases:	Should be done before update user information use case	
Stakeholders:	User Management	
Pre-conditions:	User Management subsystem must be available User must obtain permission to use the system Administrator must be granted permission to add users	
Post-conditions:	User account must be created and saved. User's private information in the system must be protected. User must allow their rules and use the system properly. Account must be associated with Asset.	
Flow of Events:	Actor	System
	1.System Administrator wants to add a new user to system.Administrator create user account and enter user's address. 2. Administrator enter other information,such as name,phone number and so on. 3. Get the detail information of the new account	1.1 System verify administrator's permission. 1.2 System check if the address already exist. 2.1 System verify the other information that administrator input. 2.2 System create an account for user. 3.1 Return the detail information of the new information.
Alternative Flow	2.1	
Exception Conditions:	1.1 The administrator has no permission. 1.2 Account already exists 2.1 The entered information is not in the correct format	

Table 2: Generate Incident Report description (Shama Praveen C3470623)

Use Case Name:	Generate Incident Report
Scenario:	Creating a detailed report of a specific fire incident.
Triggering Event:	use as follow-up to the completion of a fire incident response at periodic intervals.
Brief Description:	The system provides an incident report of a certain fire event complete with details of the event, response actions and their results.
Actors:	System Administrator, Operation Manager
Related use cases:	Generate Maintenance Report, Generate Asset Status Report
Stakeholders:	University Safety Committee, Fire Department, Building Occupants
Pre-conditions:	1. The fire incident has been recorded in the system. 2. Everything about response actions, outcomes, has been documented.
Post-conditions:	1. An incident report is prepared with necessary details. 2. A copy of the report is retained in the system for subsequent use. 3. The recipients of the report are informed.

Flow of Events:	Actor	System
	1 Operation Manager requests the generation of an incident report. 2. 3. 4.	1.1 System checks incident data availability. 1.2 If data is complete, the system retrieves incident data. 1.3 System compiles and processes the data. 1.4 System generates the incident report. 1.5 System stores the incident report. 1.6 System notifies relevant stakeholders that the report is available. 1.7 If data is incomplete, the system notifies Operation Manager to update the necessary information.
Alternative Flow	If the data is missing, then the system alerts the Operation Manager about missing data which needs to be filled in.	
Exception Conditions:	1: Reporting and documentation are inadequate, and some of the incident details might be lost or incomplete. 2: This issue arises when there is a system error while producing a report of the employee's output.	

Table3: **Monitor FIP Status (ZHENG LIANG c3488157)**

Use Case Name:	Monitor FIP Status
Scenario:	Monitor the real-time status of Fire Indicator Panel (FIP) continuously.
Triggering Event:	The system initialization process starts the monitoring of the FIP status. This process occurs automatically when the system is turned or reset. It can also be initiated by the Fire Response System Team and System Administrator team manually.
Brief Description:	This use case shows how the Fire Response System and System Administrator monitor the real-time status of Fire Indicator Panel (FIP) continuously to make sure the proper functioning. With the monitor, fire-related incidents can be detected.
Actors:	Fire Response System System Administrator
Related use cases:	Update FIP Configuration Perform FIP Diagnostics Log FIP Events Generate FIP Report
Stakeholders:	Facility Management Fire Response Team Safety Compliance Officer Building Occupants

	University Safety Committee	
Pre-conditions:	1. The FIP must be properly installed and adjusted in the building. 2. The System Administrator and Fire Response System has the necessary permissions and access rights. 3. The System and FIP can be monitored online, System Administrator and Fire Response System can access them anytime and anywhere.	
Post-conditions:	1. The FIP status should always be real-time status. 2. Any changes monitored should be logged and generated and reported.	
Flow of Events:	Actor	System
	1. System Administrator starts monitoring the FIP status upon initialization.	1.1 System operated.
		2. System receives the real-time FIP status continuously.
		3. The system displays all the real-time status on the dashboard.
	4. If the system successfully displays the FIP status on dashboard, the Fire Response System can access the dashboard of the FIP to monitor the status.	4a. If not, the system will display a communication failure message.

		5. The system detects abnormal changes of the FIP status.
		6. The system generates an alert.
		7. The system alerts the Fire Response System Team about the detected abnormally changes of the FIP status.
		6a. If the system successfully generates an alert for the Fire Response Team, it will display a fail to alert message.
	8. If the system successfully generates an alert for the Fire Response Team on the dashboard, then the Fire Response Team will make a response to take necessary actions.	
		9. System logs all the data, status updates, and the alerts.
	3a. The system meets a communication failure with the	

	FIP.	
		5a. The system meets the communication failure.
Exception Conditions:	2a. The system makes wrong analyses or can't analyse due to the bugs. 3b. The System exists connect problems. Fire Response System Team can't access the dashboard to monitor the FIP status through network. 5a. Power failure interrupts the generating of alert. 8a. The storge is full, the system can't log any updates and reports.	

Table4: Update FIP Configuration (Yin Ho Chan)

Use Case Name:	Update FIP Configuration
Scenario:	Modify the configuration settings of the Fire Indicator Panel (FIP).
Triggering Event:	The System Administrator receives a request from the FIP Technician to update the FIP configuration caused by system upgrades and newly identified issues.
Brief Description:	Involves the modification of the FIP configuration settings. The process is initiated by the System Administrator based on the input from the FIP Technician to ensure FIP operates correctly and integrates with other systems.
Actors:	System Administrator FIP Technician
Related use cases:	Monitor FIP Status, Perform FIP Diagnostics, Log FIP Events
Stakeholders:	Facility Management Fire Response Team Safety Compliance Officer
Pre-conditions:	1. The System Administrator has the necessary permissions and access rights. 2. Relevant configuration documentation has been provided by FIP Technician.
Post-conditions:	1. The FIP configuration is successfully updated. 2. Updated configuration settings are being implemented

	for The FIP to operate. 3. The updated configuration settings are being verified 4. All relevant stakeholders are being notified of the updated configuration changes.	
Flow of Events:	Actor	System
	1. The FIP Technician identifies the need to update the FIP configuration 2. The FIP Technician informs the System Administrator. 3. The System Administrator receives the updated configuration document from the FIP Technician. 4. The System Administrator initiates the FIP configuration update process. 5. The System	4.1 System prompts for authentication and authorization. 5.1 System displays the current FIP configuration

	Administrator retrieves the current configuration settings.	settings.
	6. The System Administrator modifies the configuration settings based on the FIP Technician's update document.	6.1 System validates the new configuration settings.
	7. The System Administrator confirms and saves the changes.	7. The system updates the FIP with the new configuration settings and logs the changes. 8. The System notifies all relevant stakeholders of the updated configuration changes.
	8. The System Administrator log off the system.	8.1 System exists.
Alternative Flow	6. The System Administrator encounters invalid configuration settings.	

	a. The System notifies the System Administrator of the error and provides details. b. The System Administrator consults with the FIP Technician to review and correct the configuration settings.
Exception Conditions:	4.1: Authentication fails. a. The system denies access and logs the failed attempt. b. The system will display a message to the System Administrator. c. System exits prior to display current configuration settings. 7.1: The update process is interrupted due to bad internet connection or power failure. a. The system rolls back to previous configuration and logs the incident. b. System exits prior to saving new configuration settings.

Table 5: Authenticate User Login Saurabh Dev(c3483990)

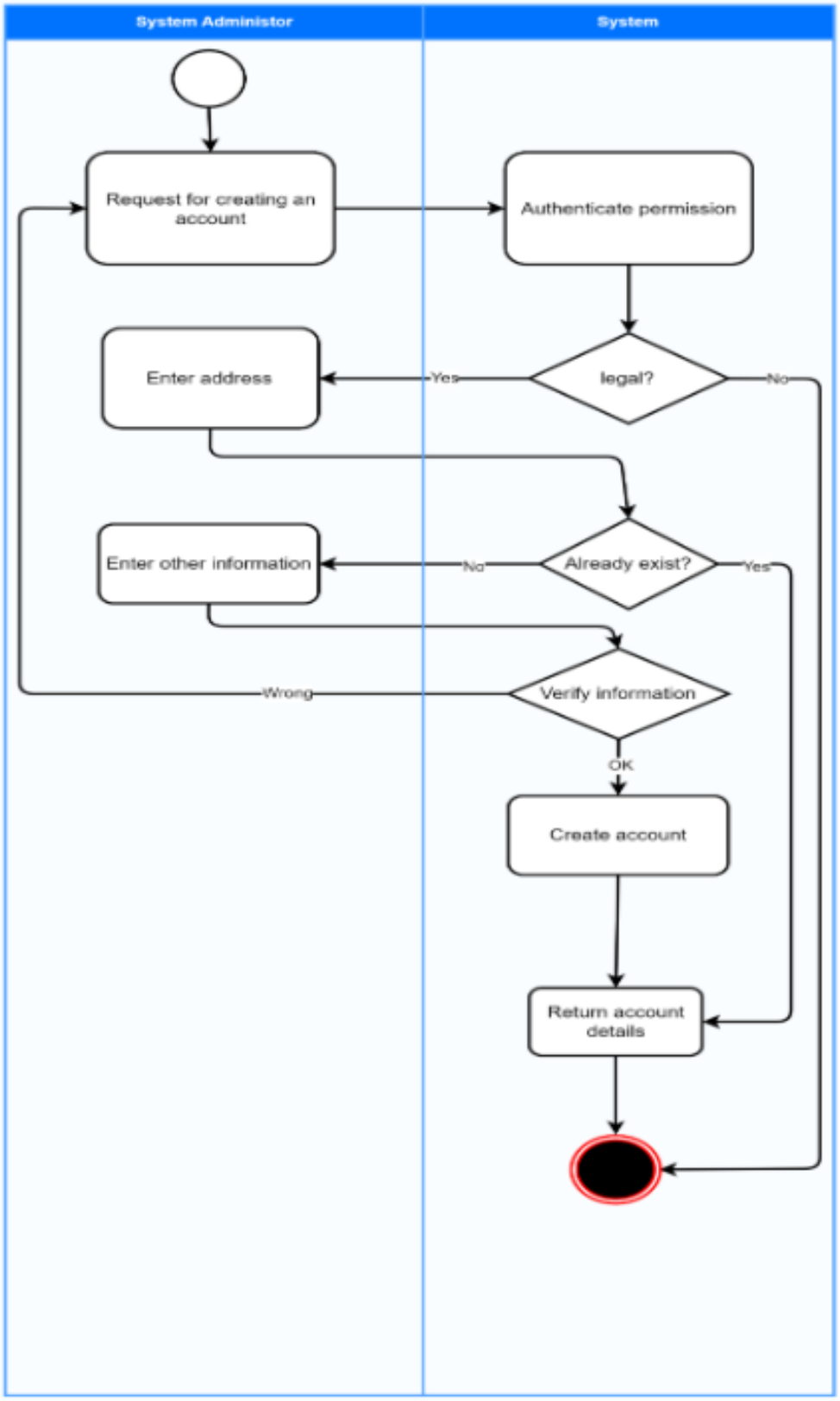
Use Case Name	Authenticate User Login	
Scenario	User attempts to enter the system	
Triggering Event	User enters the website and goes to the login page where he enters his personal details and password.	
Brief Description	Involves the user entering his or her username and password to log into the respective system. It checks the submitted details with the correct ones and allows the user access to the system if the inputs are accurate.	
Actors	User, System	
Related Use Cases	Register New user Reset Password Logout User Update User Profile	
Stakeholders	User System Administrator	
Preconditions	1. The System assumed that if a user has an account with the system, he/she is a participant. 2. They should provide valid credentials that include a username and a password that is necessary for login. 3. The system should include the log-in page.	
Postconditions	1. The client can access the system through the authentication stage. 2. If the user enters incorrect information, he is asked to input correct login particulars or try performing a password reset.	
Flow of Events	Actor	System
	1.User Navigate to the login page	
		2.Login page is displayed in system
	3.The user enter Username and password	

		4.The Credentials is captured and encrypted by the system
		5. System verifies credentials against the database.
	6. If credentials are correct, the user is redirected to the home page.	5.a. If credentials are correct, proceed to step 6.
		5.b. If credentials are incorrect, the system displays an error message.
	7. User accesses the home page.	
Alternate Flow	5.c. If the username is not found in the database The system informs the user that the username is incorrect. Return to step 2. 5.d. If the username is found but the password does not match The system informs the user that the password is incorrect. Return to step 2.	
Exception Conditions:	5.b. Account Lockout a. If the user exceeds a specified number of failed login attempts, the account is locked. b. The system provides instructions for unlocking the account. 2.System Unavailable c. If the system is down, an error message is displayed, and the user is unable to log in.	

4 Activity Diagrams

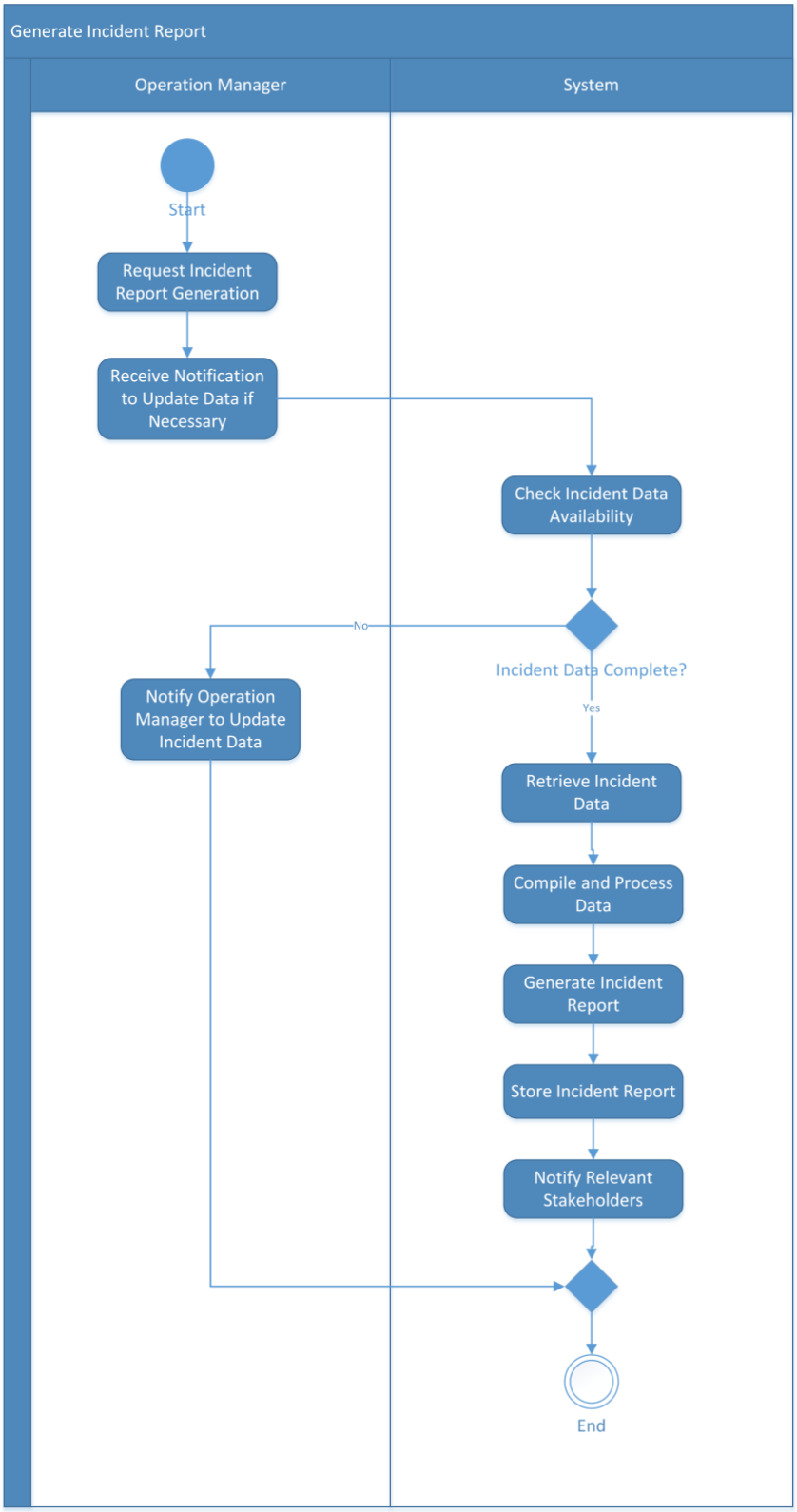
Create User Account (Kun Zhang)

The activity diagram illustrates the process of creating an account by a system administrator. The administrator begins by requesting the creation of an account and entering the necessary details such as address and other required information. The system then verifies the request, checks its validity, and confirms if the account already exists. If the information is accurate and the account does not exist, the system creates the account and provides the account details. If there are any issues during verification, the administrator is prompted to re-enter the necessary information.


System Administor
System
Request for creating an account
Authenticate permission
Enter address
Yes
legal?
No
Enter other information
No
Already exist?
Yes
Wrong
Verify information
OK
Create account
Return account details

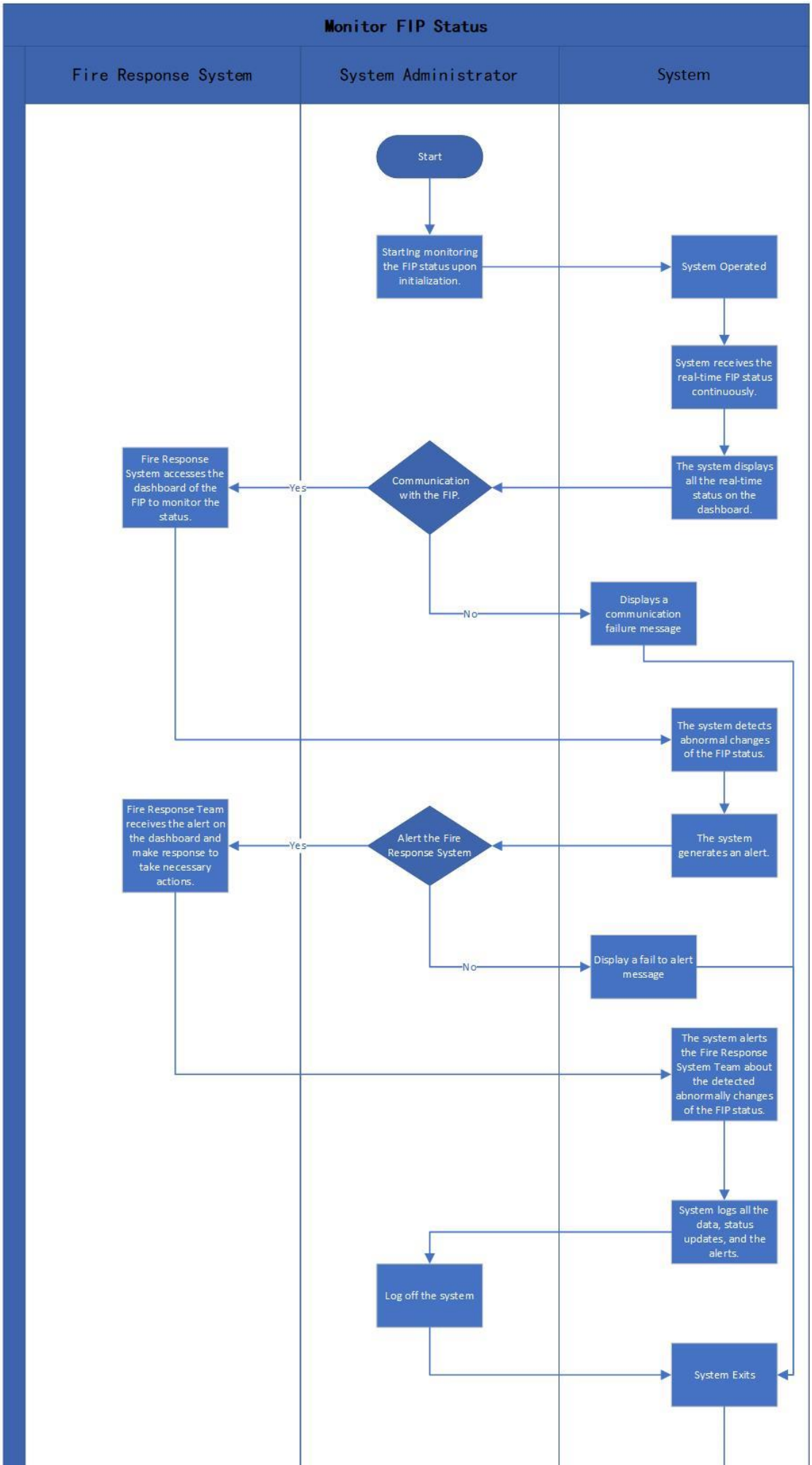
Generate Incident Report: (Shama Praveen C3470623)

The activity diagram below describes the generation of an incident report under a fire management system. The process starts when the Operation Manager orders the report. The system checks for a condition where incident data is available. In case the data is complete, the system pulls or gathers, aggregate, analyze and create the report and the report is saved and users informed. If the data is missing, a message is sent to the Operation Manager to add the data and then resume the process.


Generate Incident Report
Operation Manager
System
Start
Request Incident Report Generation
Receive Notification to Update Data if Necessary
Check Incident Data Availability
No
Incident Data Complete?
Yes
Notify Operation Manager to Update Incident Data
Retrieve Incident Data
Compile and Process Data
Generate Incident Report
Store Incident Report
Notify Relevant Stakeholders
End

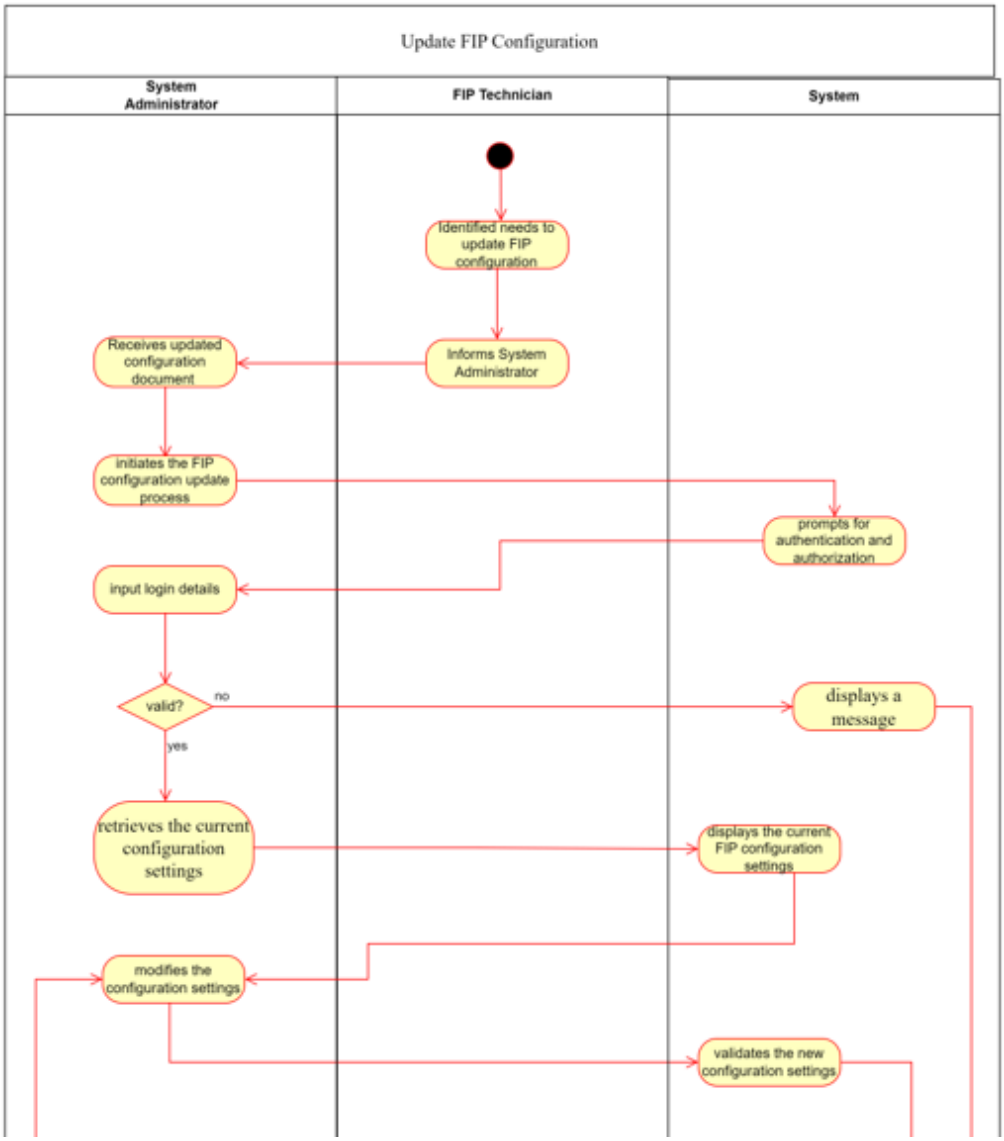
Monitor FIP Status (ZHENG LIANG c3488157)

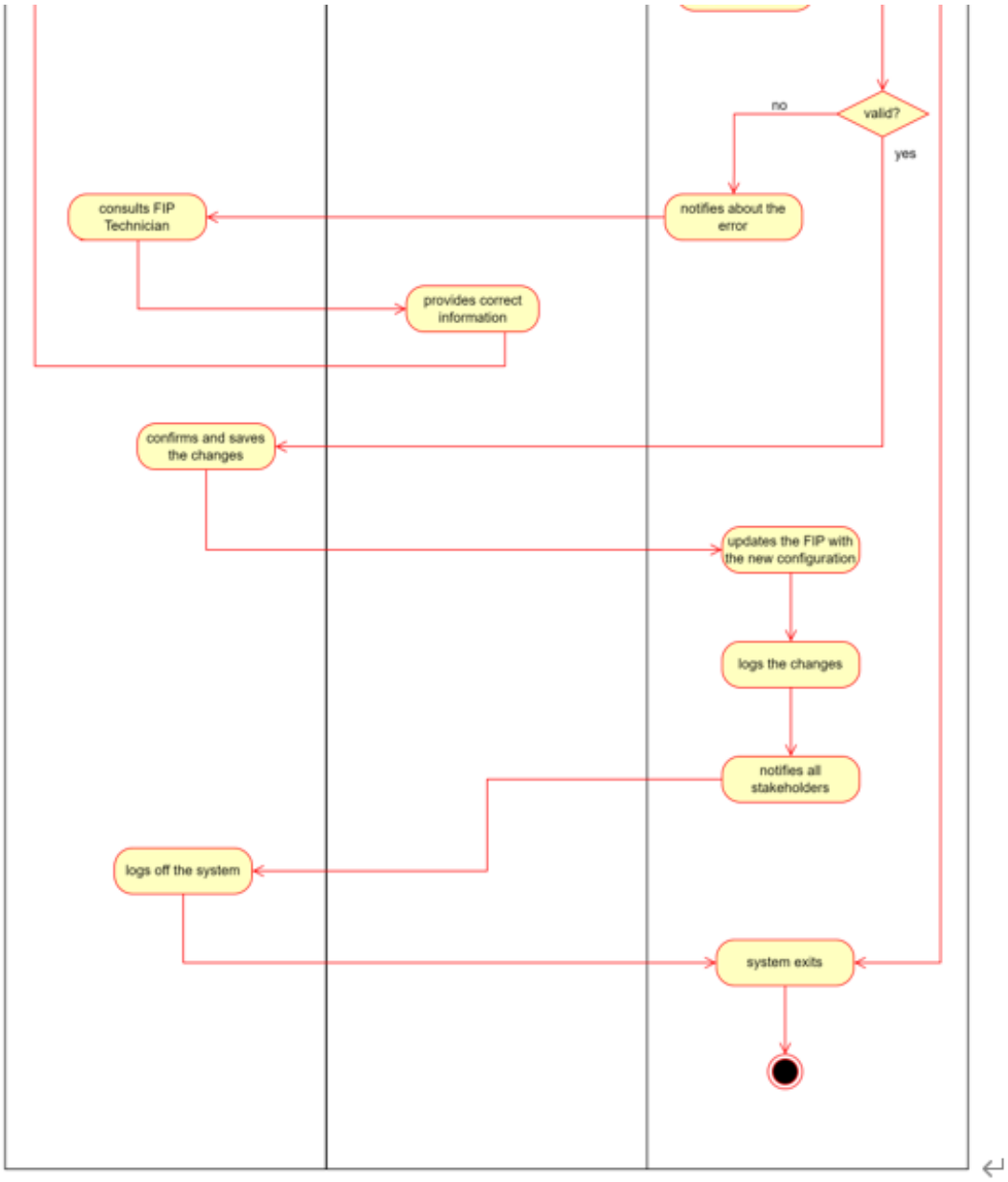
This diagram shows how the Fire Response System and System Administrator monitor the real-time status of Fire Indicator Panel (FIP) continuously to ensure the proper functioning. With the monitor, fire-related incidents can be detected. This flowchart describes the process of monitoring the real-time status of the Fire Indicator Panel (FIP). It details the roles of the System Administrator and the Fire Response System in initializing the monitoring process, displaying real-time status on a dashboard, detecting abnormal changes, generating alerts, and logging data. The chart also includes alternative flows for communication failures and alerts. The basic flow involves the System Administrator starting the monitoring, the system receiving and displaying FIP status, detecting abnormalities, generating alerts, notifying the Fire Response System Team, and logging all data and status updates.


Monitor FIP Status
Fire Response System
System Administrator
System
Start
Starting monitoring the FIP status upon initialization.
System Operated
System receives the real-time FIP status continuously.
The system displays all the real-time status on the dashboard.
Communication with the FIP.
Yes
Fire Response System accesses the dashboard of the FIP to monitor the status.
No
Displays a communication failure message
The system detects abnormal changes of the FIP status.
The system generates an alert.
Alert the Fire Response System
Yes
Fire Response Team receives the alert on the dashboard and make response to take necessary actions.
No
Display a fail to alert message
The system alerts the Fire Response System Team about the detected abnormally changes of the FIP status.
System logs all the data, status updates, and the alerts.
Log off the system
System Exits

Update FIP Configuration (Yin Ho Chan)

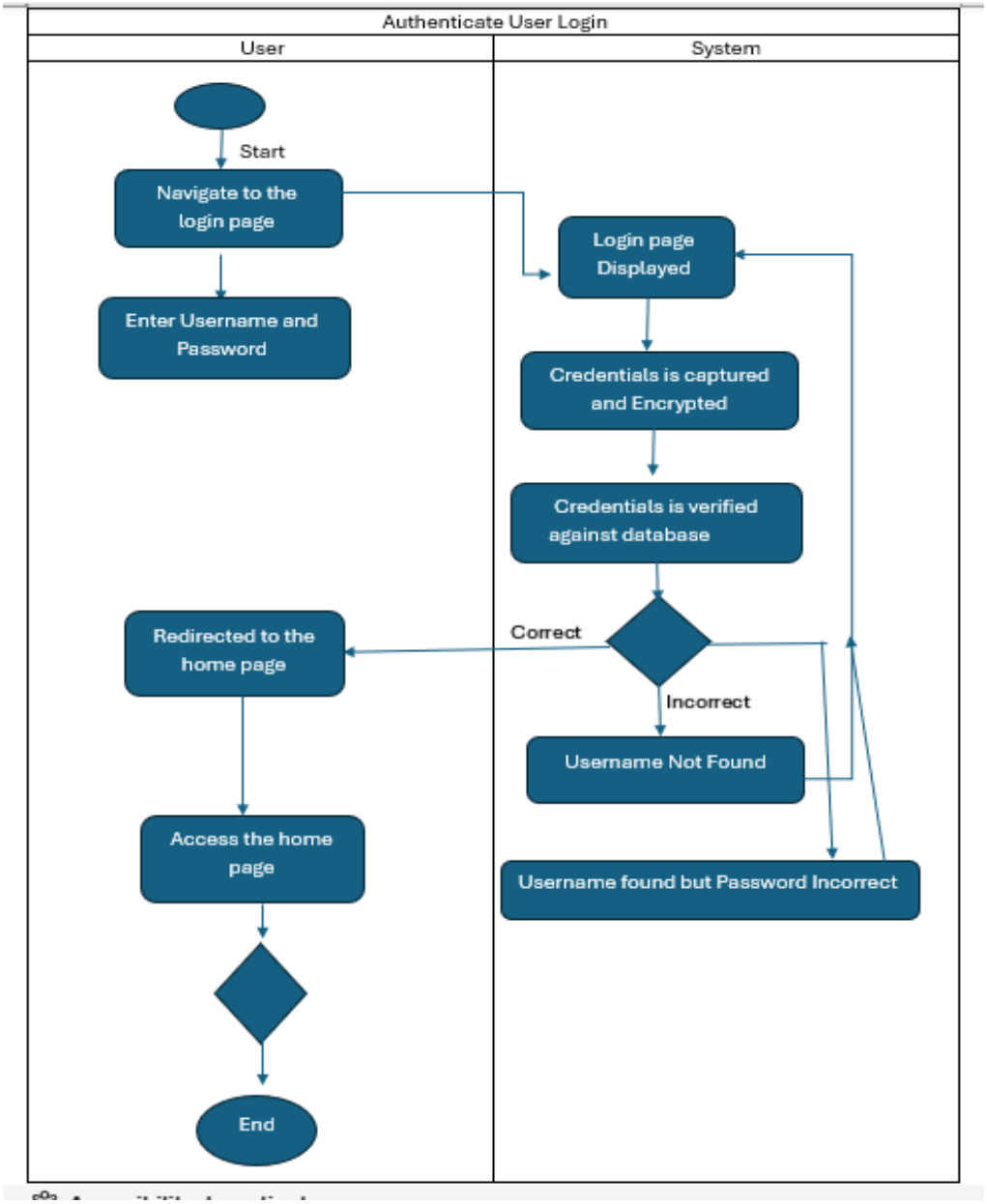
The Update FIP configuration step is crucial for maintaining the integrity and functionality of the Fire Indicator Panel (FIP) system. When updating the configuration, after FIP Technician provides the necessary changes, the current settings must be retrieved and reviewed by the System Administrator. which the updates are then implemented by the System Administrator. It is important that the new configuration settings are validated to ensure they do not introduce errors or conflicts. Once validated, the updated settings are saved, and the changes are logged. The System must notify all relevant stakeholders about the update.



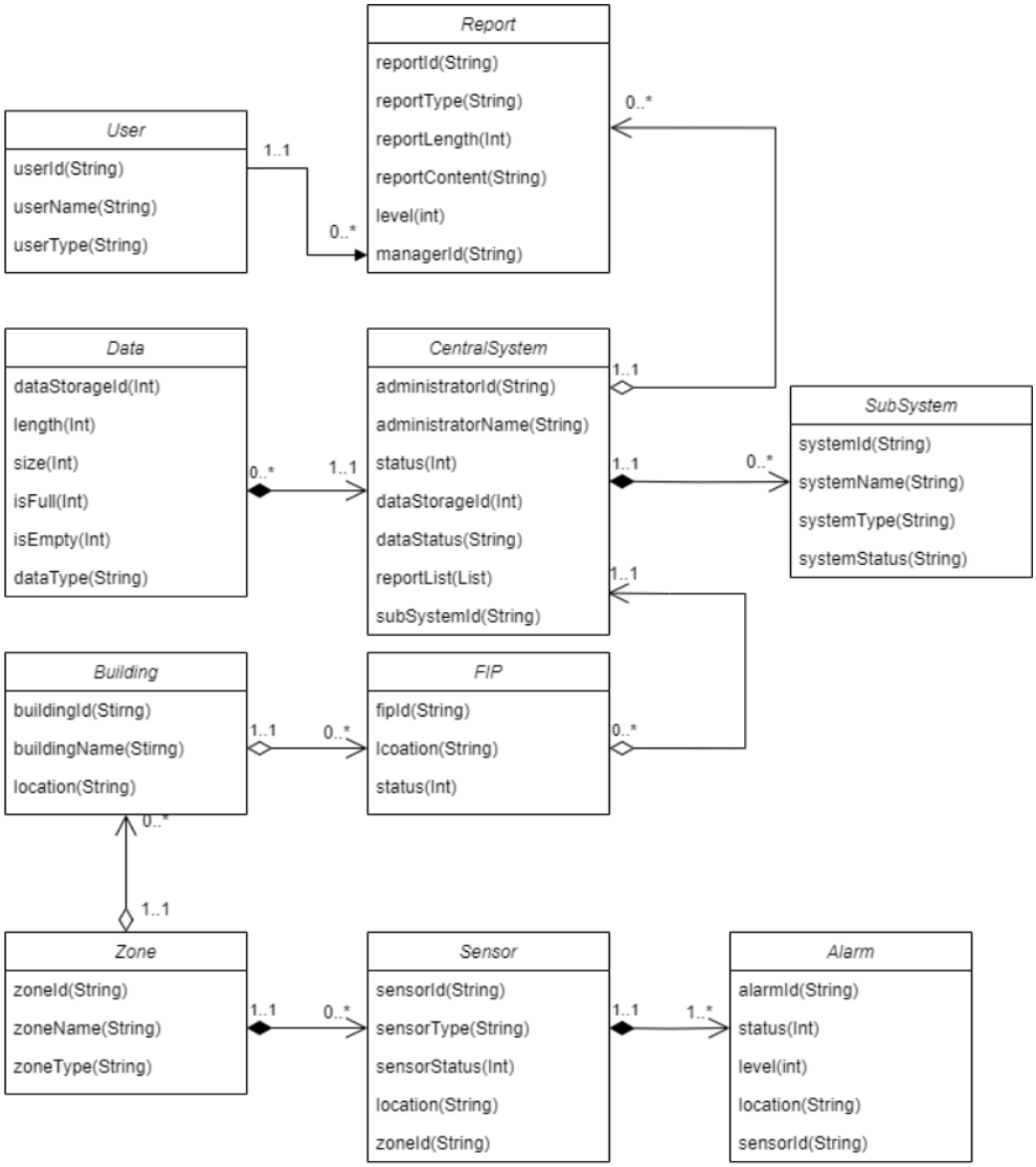

valid?
no
yes
notifies about the error
consults FIP Technician
provides correct information
confirms and saves the changes
updates the FIP with the new configuration
logs the changes
notifies all stakeholders
logs off the system
system exits

Authenticate User Login (Saurabh Dev)

Activity diagrams start from the user sessions and progress to the roaming – the system presents the user with a login form the user key then enters the username and password. These are then stored by the system and further the passwords are encrypted. The software compares what has been input by the user to the data already saved in the database. For instance, there is a decision point that a user encounters when the system asks if the credentials provided are correct. They were used in the authentic process. If they are right, the user is directed to the home page and accesses the system. On the other hand, if the credentials are wrong, a message stating that the input is invalid appears and no reading information will be provided.


Authenticate User Login
User
System
Start
Navigate to the login page
Enter Username and Password
Login page Displayed
Credentials is captured and Encrypted
Credentials is verified against database
Correct
Incorrect
Redirected to the home page
Access the home page
Username Not Found
Username found but Password Incorrect
End

5 Domain Class Diagram



6 Team Management

6.1 Meeting Minutes

Meeting 1

INFO6030 Systems Analysis and Design

Minutes of meeting

Team___Group 3_____ Place_____online__Teams Meetings_____
Date/Time___26/05/2024___12:30pm to 13:30pm

In attendance

Yin Ho Chan
Kun Zhang
Saurabh Dev

Apologies

N/A

Agenda

- Matters arising from previous meeting
  N/A
- Agenda items (as needed)
  Task from below.
- Date, time and place for next meeting
  Next Week till 2nd of June
- Matters for consideration at next meeting
  Considering each member work

Action sheet

Task	Responsible	Due	Notes
2. Introduction to the report	Saurabh	Today on 26th may	Point out the report and objectives.
3. Business rules	C=Kun D=Yin Ho G=Saurabh B=All of us	Till 29th May	10 Rules to point out for each section and for B 30 points
4. Use Case Diagram	A	Later	Discuss the use case in the next meeting.

Meeting 2

INFO6030 Systems Analysis and Design

Minutes of meeting

Team_____Group 3_____ Place____Online Team Meeting_____
Date/Time___05/06/2024_____

In attendance
Yin Ho Chan
Kun Zhang
Saurabh Dev

Apologies

N/A

Agenda

- Matters arising from previous meeting
- Agenda items (as needed)
- Date, time and place for next meeting till 12th Jun
- Matters for consideration at next meeting is that considering each member work

Action sheet

Task	Responsible	Due	Notes
Discussion about the previous work	All group members	Today	Discussion about the previous work
Combine the previous work in report	All group members	Today	Combine the previous work in report
List all use cases and decide	All group members	Till 11th Jun	List all use cases and decide

Meeting 3

INFO6030 Systems Analysis and Design

Minutes of meeting

Team__Group 3_____ Place_Online Team Meeting_____
Date/Time__12/06/2024_____

In attendance
Yin Ho Chan
Kun Zhang
Saurabh Dev
Shama Praveen
Zheng Liang

Apologies
N/A

Agenda

- Matters arising from previous meeting
- Agenda items (as needed)
- Date, time and place for next meeting till 19th June
- Matters for consideration at next meeting is that considering each member work

Action sheet

Task	Responsible	Due	Notes
Discussion about the previous work	All group members	Today	Discussion about the previous work
Decide the Use Case	All group members	Today	Decide the Use Case
Combine the work in the report	All group members	Today	Combine the work in the report

Meeting 4

INFO6030 Systems Analysis and Design

Minutes of meeting

Team____Group _3___ Place___Online Team Meeting_____
Date/Time_19/06/_____

In attendance
Yin Ho Chan
Kun Zhang
Saurabh Dev
Shama Praveen
Zheng Liang

Apologies

N/A

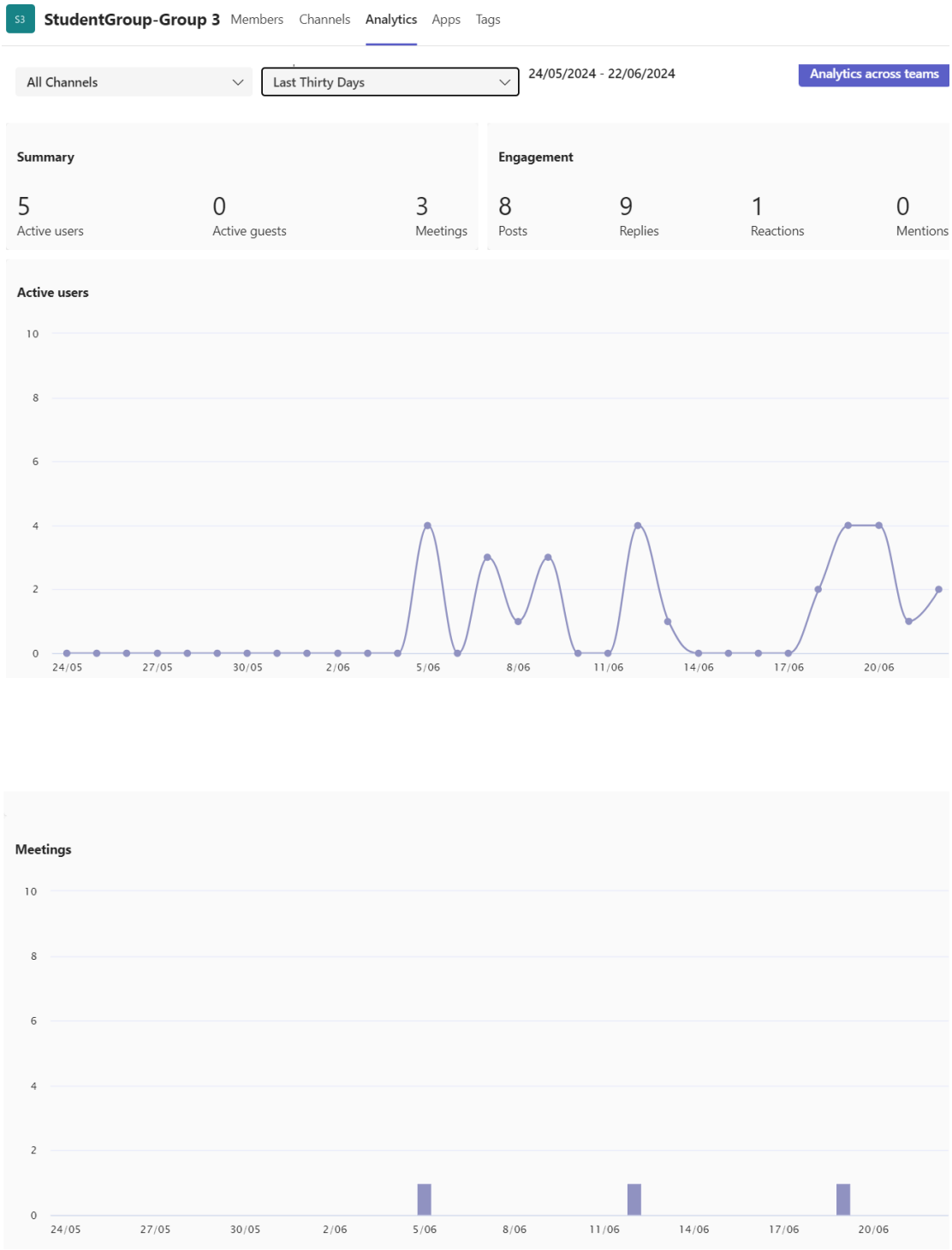
Agenda

- Matters arising from previous meeting
- Agenda items (as needed)
- Date, time and place for next meeting N/A
- Matters for consideration at next meeting N/A

Action sheet

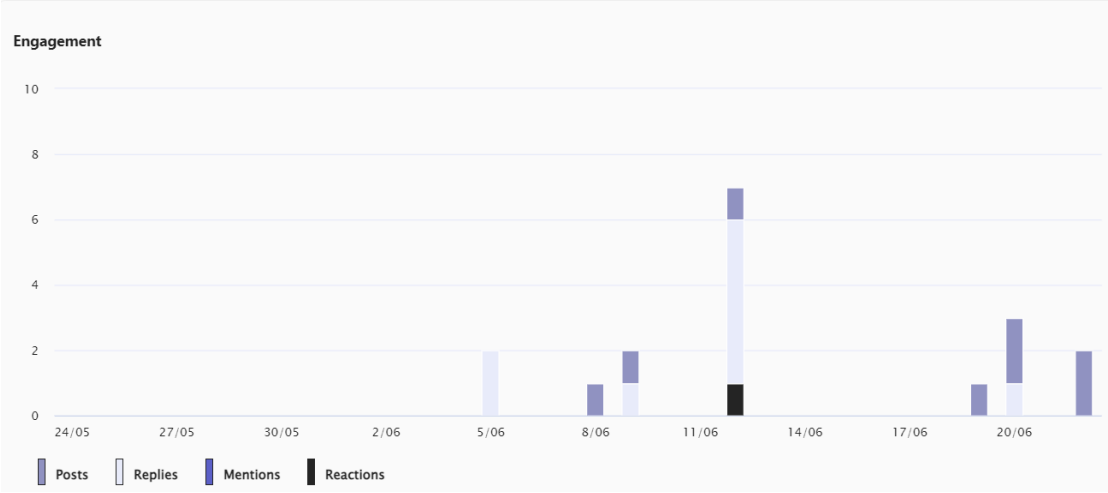
Task	Responsible	Due	Notes
Discuss the report	All group members	Today	Discuss the report
Combine the report	All group members	Today	Combine the report
Finish the Appendix	All group members	Today	Finish the Appendix

6.2 MS activity report



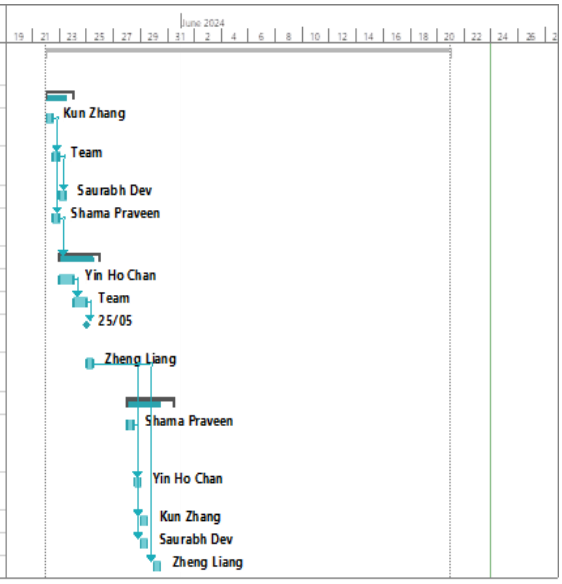
Note: The first meeting has finished on teams on the 26th of May, however it cannot be shown in the statistics due to the call is made in Teams chat room instead.

26/05 11:38 am Call ended **1h 4m 47s**

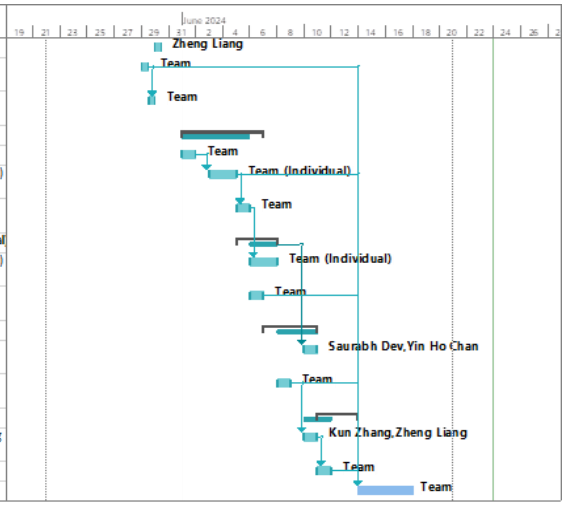


6.3 MS Project Gantt Chart

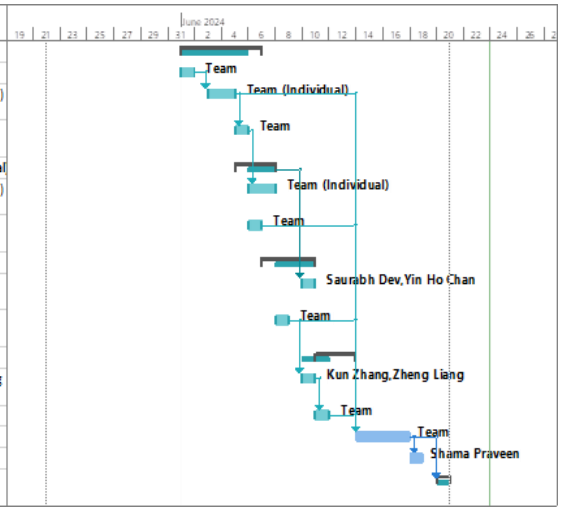
ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors	Team Member
0			**Uon fire management system**	**22 days**	**Wed 22/05/24**	**Thu 20/06/24**		
1			**1 Project Planning**	**2 days**	**Wed 22/05/24**	**Thu 23/05/24**		**Team**
2			1.1 Set Project Goals and objective	0.5 days	Wed 22/05/24	Wed 22/05/24		Kun Zhang
3			1.2 Assign Roles and Responsblity	0.5 days	Wed 22/05/24	Wed 22/05/24	2	Team
4			1.3 Communication Pla	0.5 days	Thu 23/05/24	Thu 23/05/24	3	Saurabh Dev
5			1.4 Develop Initial Project plan	0.5 days	Wed 22/05/24	Wed 22/05/24	2	Shama Praveen
6			**2 Requirement Gatherin**	**3 days**	**Thu 23/05/24**	**Sat 25/05/24**	**5**	
7			2.1 Identify Requireme	1 day	Thu 23/05/24	Thu 23/05/24		Yin Ho Chan
8			2.2 Create use case list	1 day	Fri 24/05/24	Fri 24/05/24	7	Team
9			2.3 Review and validate requirement	0 days	Sat 25/05/24	Sat 25/05/24	8	Team
10			2.4 Baseline Requirement	0.5 days	Sat 25/05/24	Sat 25/05/24		Zheng Liang
11			**3 Business rule identifica**	**3.5 days**	**Tue 28/05/24**	**Fri 31/05/24**		
12			3.1 Research Applicable Legislation and Standards	0.5 days	Tue 28/05/24	Tue 28/05/24		Shama Praveen
13			3.2 Identify Work Health and Safety	0.5 days	Tue 28/05/24	Tue 28/05/24	12	Yin Ho Chan
14			3.3 Identify Ethical Rul	0.5 days	Wed 29/05/24	Wed 29/05/24	10	Kun Zhang
15			3.4 Identify Security R	0.5 days	Wed 29/05/24	Wed 29/05/24	10	Saurabh Dev
16			3.5 Identify Privacy Ru	0.5 days	Thu 30/05/24	Thu 30/05/24	10	Zheng Liang



ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors	Team Member
16			3.5 Identify Privacy Ru	0.5 days	Thu 30/05/24	Thu 30/05/24	10	Zheng Liang
17			3.6 Document Business Rules	0.5 days	Wed 29/05/24	Wed 29/05/24	13	Team
18			3.7 Review and Validate Business	0.5 days	Wed 29/05/24	Wed 29/05/24	17	Team
19			**4 Use Case Identification**	**5 days**	**Sat 1/06/24**	**Thu 6/06/24**		
20			4.1 Identify Use Case	1 day	Sat 1/06/24	Sat 1/06/24		Team
21			4.2 Use case diagram creation	2 days	Mon 3/06/24	Tue 4/06/24	20	Team (Individual)
22			4.3 Review Use case diagram	1 day	Wed 5/06/24	Wed 5/06/24	21	Team
23			**5 Activity Diagram Creati**	**3 days**	**Wed 5/06/24**	**Fri 7/06/24**		**Team (Individual**
24			5.1 Activity Diagram Creation	2 days	Thu 6/06/24	Fri 7/06/24	22	Team (Individual)
25			5.2 Review Activity Diagram	1 day	Thu 6/06/24	Thu 6/06/24		Team
26			**6 Domain Analysis**	**2 days**	**Fri 7/06/24**	**Mon 10/06/2**		**Team**
27			6.1 Identify key classes and	1 day	Mon 10/06/24	Mon 10/06/24	23	Saurabh Dev, Yin Ho Chan
28			6.2 Document Domain analysis	1 day	Sat 8/06/24	Sat 8/06/24		Team
29			**7 Class Diagram Creation**	**3 days**	**Tue 11/06/24**	**Thu 13/06/24**		
30			7.1 Create preliminary class diagram	1 day	Mon 10/06/24	Mon 10/06/24	28	Kun Zhang, Zheng Liang
31			7.2 review Class diagra	1 day	Tue 11/06/24	Tue 11/06/24	30	Team
32			**8 Report Writing**	2 days	Fri 14/06/24	Mon 17/06/24	5,10,17,21,25,28,3	Team



ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors	Team Member
19			**4 Use Case Identification**	**5 days**	**Sat 1/06/24**	**Thu 6/06/24**		
20			4.1 Identify Use Case	1 day	Sat 1/06/24	Sat 1/06/24		Team
21			4.2 Use case diagram creation	2 days	Mon 3/06/24	Tue 4/06/24	20	Team (Individual)
22			4.3 Review Use case diagram	1 day	Wed 5/06/24	Wed 5/06/24	21	Team
23			**5 Activity Diagram Creati**	**3 days**	**Wed 5/06/24**	**Fri 7/06/24**		**Team (Individual**
24			5.1 Activity Diagram Creation	2 days	Thu 6/06/24	Fri 7/06/24	22	Team (Individual)
25			5.2 Review Activity Diagram	1 day	Thu 6/06/24	Thu 6/06/24		Team
26			**6 Domain Analysis**	**2 days**	**Fri 7/06/24**	**Mon 10/06/2**		**Team**
27			6.1 Identify key classes and	1 day	Mon 10/06/24	Mon 10/06/24	23	Saurabh Dev, Yin Ho Chan
28			6.2 Document Domain analysis	1 day	Sat 8/06/24	Sat 8/06/24		Team
29			**7 Class Diagram Creation**	**3 days**	**Tue 11/06/24**	**Thu 13/06/24**		
30			7.1 Create preliminary class diagram	1 day	Mon 10/06/24	Mon 10/06/24	28	Kun Zhang, Zheng Liang
31			7.2 review Class diagra	1 day	Tue 11/06/24	Tue 11/06/24	30	Team
32			**8 Report Writing**	2 days	Fri 14/06/24	Mon 17/06/24	5,10,17,21,25,28,3	Team
33			**9 Ms gantt Chart Creatio**	1 day	Tue 18/06/24	Tue 18/06/24	32	Shama Praveen
34			**10 Final Review and documentation**	**1 day**	**Thu 20/06/24**	**Thu 20/06/24**	**32**	**Team**



6.4 Team Pre-Action Plan

Note: Pre-Action Plan was submitted before the change of teammates, therefore it did not show the details of two new teammates.

Informing course coordinator Pre-project action plan

(Must be completed and shown in lab class Week 3)

"Whenever anyone goes to his or her associates and says: "This is *what* I am good at. This is *how* I work. These are my *values*. This is the contribution I plan to concentrate on and the results I plan to deliver" the response is *always*: "This is most helpful. But why haven't you told me *earlier*?"" (Drucker, 1999 p.187).

Course Code: INFO6030	**Course Name: System Analysis and Design**
Group Name: Group 3	

Name (as in university records)	Student Number	UoN E-Mail	Other Contact Type (Mobile etc.)
Kunzhang	C3444056	C3444056@uon.edu.au	0430047046
Chan Yin Ho	C3459276	C3459276@uon.edu.au	0433807550
Saurabh Dev	C3483990	C3483990@uon.edu.au	0452647323

Discussion and agreement made of the grade expectations of all group members	Yes
MS Teams site created by lab w2	Yes
1st Meeting completed by lab w2	Yes

Project Action Plan

Event	Agreed Action
Team member not participating in discussions regularly	Action 1 Asking for the reason for not attending. Action 2 Giving warning through email. Action 3 Informing course coordinator.
Team member not completing work	Action 1 Asking for the reason for not completing. Action 2 Giving the deadline for submission with the latest work given to him.

Event	Agreed Action
	Action 3 Informing course coordinator
Team member dropping out	Action 1: Contact lab Academic. Action 2: Possible adjustment of tasks Action 3: N/A
Discussions not achieving goals	Action 1 Need to find the exact issue where we are not. Action 2 Discuss the plan to resolve the issue. Action 3 Finding the exact solution so that same problem won't occurred again.
Conflict between members (please include as many areas of conflict you foresee)	Action 1 Try to respect and listen to other opinions. Action 2 List out all conflict points and vote. Action 3 Let team leader make final decision.
Member not contributing in a timely manner	Action 1 Informing about the task to be done in timely manner. Action 2 Giving the final deadline to get the work done given to him. Action 3 Informing the course coordinator
Member dominating group	Action 1 Informing him personally. Action 2 Coordinate with the team about the member on what to do next. Action 3

Event	Agreed Action
	Informing the course coordinator

Who will do what in this group work?

Task	Person(s) responsible
2,3(c, g),7(a, b, c)	Saurabh Dev
3.d,4,8	Chan Yin Ho
5,6	Kunzhang

Date shown in Lab Class 2024/05/29 Date 2024/06/01

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Drucker, P. F. (1999) Managing oneself, In *Management challenges for the 21st century*, Butterworth-Heinemann, Oxford, pp. 163-195.

7 Conclusion

Our team was efficient in identifying and recording the specifications.

Specific organizational business rules applicable to the setting were mentioned and enumerated under Work Health and Safety, Ethical, Security, and Privacy rules.
Therefore, subsystem has been created to identify all the uses cases to be included in the system as well as how they will interface.
For some of the use cases, activity diagrams were developed to describe the Workflow present within the system.
Domain analysis was done by use of class diagram to show the class name, attributes as well as associations.
The management of the team was good and activities like taking notes of the meetings, MS Teams single day reports along with Gantt Charts for tracking the progress were well managed.

Although the basic needs and the assessments were made, there are several measures and tests that could be made if there were more time and money: more detailed descriptions of use cases, and additional functions that should be incorporated into the system.

Stakeholder interviews, business specifications gathering, and preliminary systems review were completed as the first step before the project team defined the project.
We have created use case diagrams, activity diagrams, and even developed domain analysis in order to better understand and illustrate the system functionalities and the corresponding processes.
Workflows were properly managed to allow the team to meet its goals on time and within the stipulated timeframe.

Suggestion For Client:

It is suggested that the client should go through all the system requirements and all the analyses for the certain compliance with the University's fire safety policies and regulations.

It is recommended that a small-scale trial of the system should be run, to expose it to a limited number of users to determine the challenges that are likely to arise when the complete system will be set up.

As with all new systems, the users of Access should be trained on how to utilize the different features and capabilities that are available to them.

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